Final Project: 3D N-Body Simulations and the Zel'dovich Approximation

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1 Introduction

Numerical simulations are a central tool in modern cosmology. Hydrodynamical simulations, which model the baryonic component in addition to dark matter, can be used to study the formation of individual galaxies and their internal astrophysical processes and compare them directly to observations. In contrast, pure dark matter N-body simulations isolate gravitational dynamics and provide the theoretical basis against which more detailed, physics-rich simulations can be interpreted. Structure in the Universe grows hierarchically through gravitational instability, with density perturbations collapsing on increasingly larger scales over cosmic time [Springel et al., 2005, Planck Collaboration et al., 2020, Alam et al., 2021]. Gravity dominates this growth on large scales, making simulations crucial for testing the ΛCDM paradigm, which is quantitatively compatible with observations across non-linear scales [Vogelsberger et al., 2019].

Simulations also serve as controlled experiments to isolate gravitational physics, test analytic models and explore idealized initial conditions [Bagla, 2004]. They are essential for interpreting observations, evaluating physical processes, and constructing mock catalogs to quantify survey errors and calibrate analysis pipelines [Bagla, 2004, Angulo and Hahn, 2022]. Additionally, semi-analytic models can be included in dark-matter-only simulations as a computationally efficient alternative to full hydrodynamics [Somerville and Davé, 2015].

To model this evolution, N-body simulations solve the cosmological Poisson equation to compute particle accelerations. Updating particle states requires a time-integration scheme. While the simple forward Euler method leads to secular energy drift, the leapfrog integrator is symplectic, time-reversible, and conserves energy over long timescales without systematic drifts [Dehnen and Read, 2011]. More sophisticated schemes include variable-timestep leapfrog variants [Quinn et al., 1997] and higher-order integrators like Bullfrog, which improve accuracy in the mildly non-linear regime [Rampf et al., 2025].

Since full N-body simulations are computationally expensive, cheaper analytic

approximations are commonly used to model structure growth while the density field remains in the linear regime, where full gravitational force calculations are not yet required to accurately capture the dynamics. The most common such approach is first-order Lagrangian perturbation theory, known as the Zel'dovich approximation (ZA) [Zel'dovich, 1970]. By linearly displacing particles, the ZA remains accurate until shell-crossing, allowing analytic evolution up to an intermediate scale factor before starting the N-body solver. This significantly reduces computational cost. However, initial conditions generated via the ZA carry inaccuracies that can dominate systematic errors in non-linear predictions [Crocce et al., 2006, Michaux et al., 2020], motivating higher-order schemes.

In this work, a simple N-body solver is implemented using Zel'dovich-approximation initial conditions and a symplectic leapfrog integrator with gravitational acceleration computed via the non-uniform fast Fourier transform (NUFFT). The goal is to analyze the difference between the full N-body evolution and the pure Zel'dovich-approximation evolution of a Gaussian random field over time and to identify the scales and epochs at which the Zel'dovich approximation breaks down and the full non-linear dynamics become necessary to correctly capture structure growth.

The report is structured as follows: section 2 introduces the code and simulation setup, section 3 presents performance and validation checks against results from the lecture notes to ensure the code functions as intended, section 4 compares the full N-body simulation to the pure Zel'dovich approximation qualitatively through visualizations and quantitatively via the standard deviation of the density field and its power spectrum and finally, section 5 provides a short conclusion and outlook.

2 Setting up a 3D N-body simulation

The following section introduces the different functions used to set up the initial conditions and the simulation, their purpose and their interconnections.

2.1 Initial Conditions

For the initial conditions, a 3D Gaussian random field with a specified power spectrum was generated using `generate_gaussian_field(N, L, alpha=1.5, amplitude=5, seed=42)`, which takes the number of grid points per dimension N, the box size L, the power-spectrum slope α, the power spectrum aplitude and a random seed as input. The power spectrum amplitude was chosen empirically, through trial and error and visual inspection of the resulting density fields, to ensure visible growth of non-linear structure within a computationally feasible range of the scale factor a.

After constructing the wavevector grid, the function calls `calc_powerspectrum(k, alpha, amplitude)`, which returns the power spectrum

$$P_\delta(k) = Ak^{-\alpha}, \tag{1}$$

A complex white-noise field is then drawn, with real and imaginary parts independently normal-distributed, and the Gaussian density field is constructed in Fourier space as

$$\hat{\delta}^{(\mathrm{ini})}(\mathbf{k}) = \sqrt{Ak^{-\alpha}} \cdot \frac{\mathrm{noise}(\mathbf{k})}{\sqrt{L^3/N^3}}. \tag{2}$$

Furthermore, nine additional fields are computed, which are needed for the Zel'dovich displacement and for the shell-crossing analysis discussed below; the first derivatives $\partial_x\phi^{(\mathrm{ini})}$, $\partial_y\phi^{(\mathrm{ini})}$, $\partial_z\phi^{(\mathrm{ini})}$ and the second derivatives $\partial_{xx}\phi^{(\mathrm{ini})}$, $\partial_{yy}\phi^{(\mathrm{ini})}$, $\partial_{zz}\phi^{(\mathrm{ini})}$, $\partial_{xy}\phi^{(\mathrm{ini})}$, $\partial_{xz}\phi^{(\mathrm{ini})}$, $\partial_{yz}\phi^{(\mathrm{ini})}$ of the initial gravitational potential $\phi^{(\mathrm{ini})}$, defined via the Poisson equation $\nabla^2\phi^{(\mathrm{ini})} = \delta^{(\mathrm{ini})}$. In Fourier space, these are computed as

$$\partial_x\phi^{(\mathrm{ini})} = \mathcal{F}^{-1}\left[ik_x\frac{\hat{\delta}^{(\mathrm{ini})}(\mathbf{k})}{-k^2}\right] \tag{3}$$

$$\partial_y\phi^{(\mathrm{ini})} = \mathcal{F}^{-1}\left[ik_y\frac{\hat{\delta}^{(\mathrm{ini})}(\mathbf{k})}{-k^2}\right] \tag{4}$$

$$\partial_z\phi^{(\mathrm{ini})} = \mathcal{F}^{-1}\left[ik_z\frac{\hat{\delta}^{(\mathrm{ini})}(\mathbf{k})}{-k^2}\right] \tag{5}$$

$$\partial_{xx}\phi^{(\mathrm{ini})} = \mathcal{F}^{-1}\left[-k_x^2\frac{\hat{\delta}^{(\mathrm{ini})}(\mathbf{k})}{-k^2}\right] \tag{6}$$

$$\partial_{yy}\phi^{(\mathrm{ini})} = \mathcal{F}^{-1}\left[-k_y^2\frac{\hat{\delta}^{(\mathrm{ini})}(\mathbf{k})}{-k^2}\right] \tag{7}$$

$$\partial_{zz}\phi^{(\mathrm{ini})} = \mathcal{F}^{-1}\left[-k_z^2\frac{\hat{\delta}^{(\mathrm{ini})}(\mathbf{k})}{-k^2}\right] \tag{8}$$

$$\partial_{xy}\phi^{(\mathrm{ini})} = \mathcal{F}^{-1}\left[-k_xk_y\frac{\hat{\delta}^{(\mathrm{ini})}(\mathbf{k})}{-k^2}\right] \tag{9}$$

$$\partial_{xz}\phi^{(\mathrm{ini})} = \mathcal{F}^{-1}\left[-k_xk_z\frac{\hat{\delta}^{(\mathrm{ini})}(\mathbf{k})}{-k^2}\right] \tag{10}$$

$$\partial_{yz}\phi^{(\mathrm{ini})} = \mathcal{F}^{-1}\left[-k_yk_z\frac{\hat{\delta}^{(\mathrm{ini})}(\mathbf{k})}{-k^2}\right]. \tag{11}$$

The function returns all ten fields transformed back to real space via `numpy.fft.ifftn`, with an overall normalization factor of $(2\pi)^{3/2}$ applied to each field for consistency.

2.2 Simulation

The full N-body framework is implemented in `run_simulation(dphi, d2phi, N, L, a_end, dt_target=0.01, type="leapfrog", a_search_max=5)`, which takes the first and second derivative fields of $\phi^{(\mathrm{ini})}$ described above, the number of grid points per dimension N, the box size L and the target scale factor a_{end} at which the simulation is to be evaluated. The `type` argument selects the integration scheme, defaulting to `"leapfrog"` but also accepting `"zeldovich"` for a pure Zel'dovich-approximation evolution with no gravitational feedback. The target step size in superconformal time $\hat{t}$ is set to $d\hat{t} = 0.01$ by default and the maximum scale factor searched when determining the shell-crossing time is $a = 5$.

For simplicity, the simulation assumes an Einstein-de Sitter cosmology, with $\Omega_m = 1$ and $\Omega_\Lambda = \Omega_r = 0$. This allows the integration to be carried out in superconformal time $\hat{t}$, defined by

$$\hat{t} = 2 - \frac{2}{\sqrt{a}}, \tag{12}$$

following Equation 7.56 of the lecture notes, such that $a = 1$ corresponds to $\hat{t} = 0$. In this variable, the growth factor is equal to the scale factor, $D_+(\hat{t}) = a(\hat{t})$, which is used throughout below.

First, particles are placed on a regular Lagrangian grid. To reduce computational cost, particle positions are evolved using a linear displacement while the system is in the linear regime. Here, the Zel'dovich approximation (ZA), is used. The Zel'dovich map is given by

$$\mathbf{x}(\mathbf{q}, D) = \mathbf{q} + D\,\Psi(\mathbf{q}), \tag{13}$$

where particles are displaced from their initial position $\mathbf{q}$ according to the displacement field Ψ. In general the growing mode D_+ is used as the time variable but as mentioned above, for an Einstein-de Sitter cosmology it can be swapped with the scale factor, as $a(\hat{t}) = D_+(\hat{t})$.

2.2.1 Shell-Crossing Time

Since the linear approximation is only valid up to the time of first shell-crossing a^*, this time must be computed before switching to the full N-body integration. This is done by calling `calculate_shellcrossing(d2phi, a_search_max, da=0.01)`. In general, the time of first shell-crossing D_* is given by

$$D_* := \inf\{D > 0 : \det(\partial \mathbf{x}/\partial \mathbf{q}) = 0\}, \tag{14}$$

where the Jacobian of the Zel'dovich map follows directly from differentiating Equation 13 with respect to $\mathbf{q}$,

$$J_{ij} = \frac{\partial \mathbf{x}_i}{\partial \mathbf{q}_j} = \delta_{ij} - D\,\frac{\partial^2 \phi^{(\mathrm{ini})}}{\partial \mathbf{q}_i \partial \mathbf{q}_j}. \tag{15}$$

This results in the three-dimensional Jacobian matrix at every grid point,

$$J = \frac{\partial \mathbf{x}}{\partial \mathbf{q}} = \begin{bmatrix} J_{xx} & J_{xy} & J_{xz} \\ J_{xy} & J_{yy} & J_{yz} \\ J_{xz} & J_{yz} & J_{zz} \end{bmatrix} = \begin{bmatrix} 1 - D\,\partial_{xx}\phi^{(\mathrm{ini})} & -D\,\partial_{xy}\phi^{(\mathrm{ini})} & -D\,\partial_{xz}\phi^{(\mathrm{ini})} \\ -D\,\partial_{xy}\phi^{(\mathrm{ini})} & 1 - D\,\partial_{yy}\phi^{(\mathrm{ini})} & -D\,\partial_{yz}\phi^{(\mathrm{ini})} \\ -D\,\partial_{xz}\phi^{(\mathrm{ini})} & -D\,\partial_{yz}\phi^{(\mathrm{ini})} & 1 - D\,\partial_{zz}\phi^{(\mathrm{ini})} \end{bmatrix}. \tag{16}$$

Taking the determinant gives the scalar field

$$\det(J) = \det(\partial \mathbf{x}/\partial \mathbf{q}) = J_{xx}(J_{yy}J_{zz} - J_{yz}^2) - J_{xy}(J_{xy}J_{zz} - J_{yz}J_{xz}) + J_{xz}(J_{xy}J_{yz} - J_{yy}J_{xz}). \tag{17}$$

Shell-crossing occurs where $\det(\partial \mathbf{x}/\partial \mathbf{q})$ vanishes at some grid point $\mathbf{q}$, indicating that the Zel'dovich map is no longer locally invertible there and that the corresponding fluid element has undergone infinite compression. To estimate the first time of shell-crossing numerically, $\min_{\mathbf{q}} \det(J)$ is tracked over the grid for increasing a in steps of $da = 0.01$, up to a maximum of $a_{\mathrm{search,max}} = 5$. The function returns the first value of a at which $\min_{\mathbf{q}} \det(J) \leq 0$.

2.2.2 Switching from ZA to Leapfrog

The scale factor at which the simulation switches from ZA to the full leapfrog integration is set to $a_{\mathrm{switch}} = 0.5\,a^*$. This was chosen to ensure that the particle distribution is still smooth and well-described by linear theory when the leapfrog integrator takes over.

If `type="zeldovich"` or if a_{switch} is larger a_{end}, the particles are displaced directly to a_{end} using `displace_particles(a_end, d_phi_x, d_phi_y, d_phi_z, particles_grid_x, particles_grid_y, particles_grid_z, L)`. This function takes the first derivatives of the initial potential field, the initial grid positions and the box size as input and displaces the particles according to Equation 13. The resulting positions are wrapped modulo L to remain inside the periodic box.

Following the proof of Proposition 7.3 and Equation 5.23 of the lecture notes, the corresponding velocity is

$$\frac{dX}{d\hat{t}} = \frac{dX}{da}\,\frac{da}{d\hat{t}} = -\nabla_q \phi^{(\mathrm{ini})} \cdot a^{3/2}. \tag{18}$$

This is evaluated for each axis at $a = a_{\mathrm{end}}$. The displaced positions and corresponding velocities are then returned by `run_simulation()`.

If instead `type="leapfrog"` and $a_{\text{switch}} < a_{\text{end}}$, the particles are displaced only up to a_{switch}. These positions and the corresponding computed velocities then serve as the initial conditions for the leapfrog integrator. Starting from a_{switch}, the following loop runs until a_{end} is reached: the step size in terms of the scale factor da is computed as $dt_{\text{target}} \cdot a^{1.5}$, the next scale factor is evaluated as $\min(a + da, a_{\text{end}})$ and the function `step(X, V, a, da, a_next, N, L)` performs one leapfrog step. Scaling da by $a^{1.5}$ compensates for the fact that a fixed step in a corresponds to an increasingly large step in superconformal time $\hat{t}$ as $a \to 0$, which would otherwise leave the integration poorly resolved shortly after the switch.

2.2.3 The Leapfrog Step

Specifically, `step()` performs one leapfrog step, following a symmetric drift-kick-drift (DKD) scheme

$$X(\hat{t} + \varepsilon/2) = X(\hat{t}) + \tfrac{\varepsilon}{2} V(\hat{t}) \tag{19}$$

$$V(\hat{t} + \varepsilon) = V(\hat{t}) + \varepsilon \, \tfrac{3}{2} a(\hat{t}_{\text{mid}}) \, \mathbf{A}\big(X(\hat{t} + \varepsilon/2)\big) \tag{20}$$

$$X(\hat{t} + \varepsilon) = X(\hat{t} + \varepsilon/2) + \tfrac{\varepsilon}{2} V(\hat{t} + \varepsilon), \tag{21}$$

where $\varepsilon = dt = \hat{t}_{\text{end}} - \hat{t}_{\text{begin}}$ is the step size in superconformal time, computed using Equation 12, $a(\hat{t}_{\text{mid}})$ is the scale factor evaluated at the temporal midpoint of the step and the prefactor $3/2$ arises from the Einstein-de Sitter Poisson equation, $\nabla^2 \phi = \frac{3}{2a} \delta$ (Equation 7.1b from the lecture notes with $\Omega_m = 1$).

The gravitational acceleration $\mathbf{A} = -\nabla \phi$ is obtained via `calc_acc_nufft(X[:,0]/L, X[:,1]/L, X[:,2]/L, N)`. This function uses `finufft.nufft3d` to solve the Poisson equation in Fourier space, where the spatial derivative becomes algebraic:

$$\hat{A}_i(\mathbf{k}) = -i k_i \hat{\phi}(\mathbf{k}) = -i \frac{k_i}{|\mathbf{k}|^2} \hat{\delta}(\mathbf{k}). \tag{22}$$

The function returns the real-space components of the acceleration field, which is then used in Equation 20.

Finally, `step()` wraps the updated positions modulo L, advances a to $a + da$ and returns the updated positions, velocities, scale factor and acceleration. `run_simulation()` uses these to initialize the next step. Once the updated scale factor reaches or exceeds a_{end}, the loop terminates and the final positions, velocities and acceleration are returned.

3 Validation and Performance Checks

Before running higher-resolution 3D simulations, several checks were performed to compare against the lecture notes and to ensure correct performance.

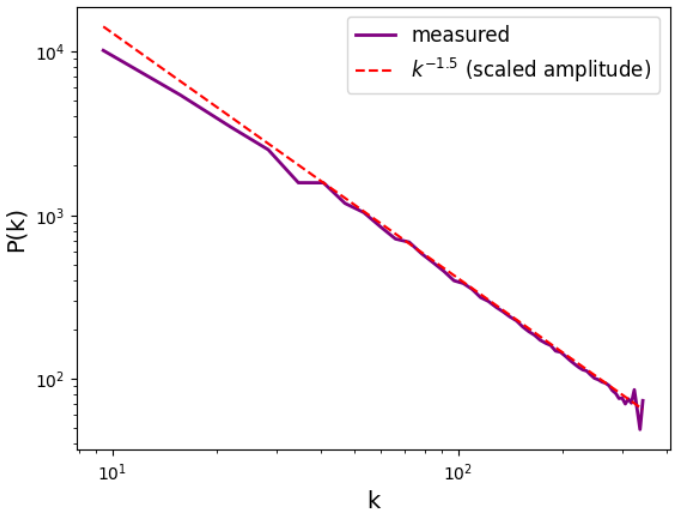


Figure 1: Measured powerspectrum (purple) of the initial field with N=64, L=1 and $\alpha = 1.5$ compared to $k^{-1.5}$ (red), with matched amplitude.

3.1 Validating the Power Spectrum

First, it was investigated whether `generate_gaussian_field` returns an initial field that encodes the desired power spectrum. For this, `compute_powerspectrum3d(field, L, N)` was used, which computes the power spectrum of a given field; it sets up the wavevector grid, Fourier transforms the input field, computes the power spectrum as $P(k) = |\hat{\delta}(\mathbf{k})|^2/N^3$, bins it azimuthally in $|\mathbf{k}|$, and returns the binned power spectrum together with the corresponding wavenumbers. This was computed for a field with $L = 1$, $N = 64$, $\alpha = 1.5$ and amplitude $A = 25$. For comparison, $k^{-1.5}$ was computed and its amplitude rescaled to match the measured spectrum. Figure 1 shows good agreement at small scales, with deviation appearing at large scales, most likely due to the limited number of independent Fourier modes available at low k in a finite box. The slope agreement provides confidence that `generate_gaussian_field()` correctly encodes the intended power-law power spectrum, though the absolute amplitude normalization was not verified independently.

3.2 1D Plane Wave Evolution

For a one-dimensional plane wave, the simulation should produce the expected patterns in x-v phase space for both the ZA and leapfrog integrators. To test this, one-dimensional initial conditions were set up following

$$\Psi_x(\mathbf{q}) = \frac{1}{2\pi} \sin\big(2\pi(\mathbf{q} - 0.25)\big), \tag{23}$$

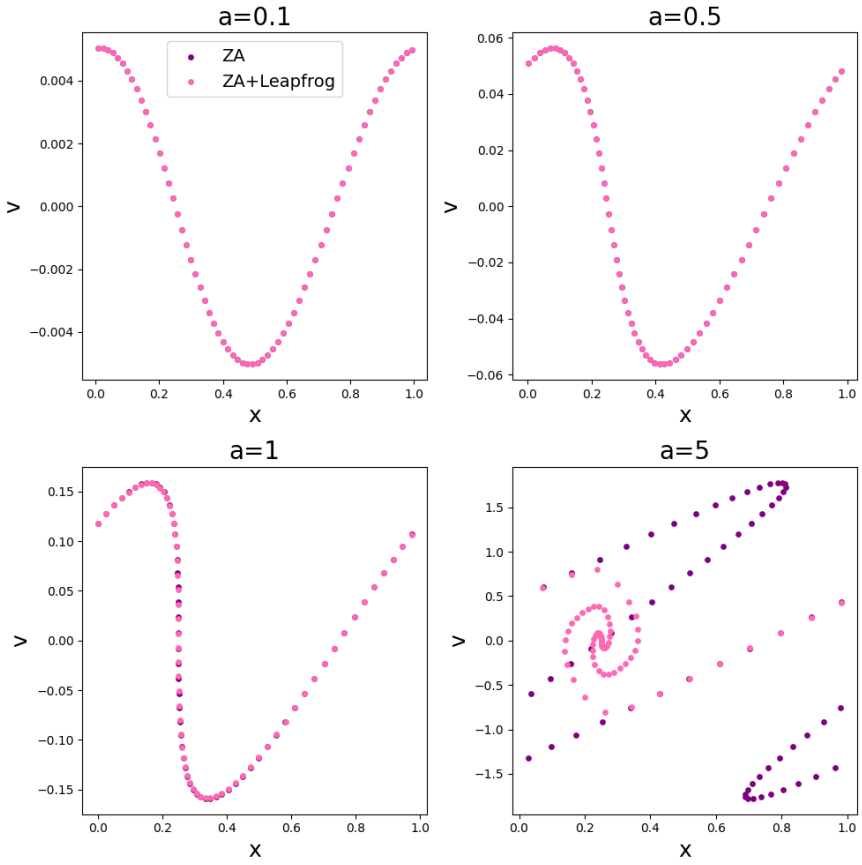


Figure 2: Evolution of a single-mode plane wave evolved using ZA (purple) and the ZA+Leapfrog combination (pink). The panels show the evolution with the scale factor; $a \in \{0.1, 0.5, 1, 5\}$.

which is similar to the plane wave described by Equation 7.57 of the lecture notes but shifted by a quarter-period. This was chosen so that the resulting caustic forms at the center of the periodic box rather than at its boundary, for clearer visualization, however, this shift does not affect the collapse dynamics or shell-crossing time. `run_simulation()` was then called with $L = 1$, $N = 64$, and varying $a_{\text{end}} \in \{0.1, 0.5, 1, 5\}$, for both `type="zeldovich"` and `type="leapfrog"`. Figure 2 shows the results.

With this configuration, shell-crossing occurs at $a^* = 1.01$ and leapfrog integration begins at $a_{\text{switch}} = 0.505$. In the upper two panels, the ZA and leapfrog solutions overlap perfectly, since shell-crossing has not yet occurred and prior to a_{switch}, the leapfrog solution is computed via linear displacement, so the two integrators are identical up to that point by construction. The near-sinusoidal

shape of the displacement field is already beginning to visibly distort by $a = 0.5$, which is a sign of the approach to shell-crossing. In the third panel, at $a = 1$, the leapfrog integrator has taken 82 steps and the two solutions have begun to diverge. By $a = 5$, the ZA solution has distorted further into steep, folded diagonals, while the leapfrog integrator has taken 194 steps by this point and has wound into the characteristic spiral shape in phase space.

This evolution is expected and a meaningful validation of the integrator. The Zel'dovich approximation is a non-interacting map. Once particles reach the caustic, ZA has no mechanism to represent any gravitational feedback and simply continues extrapolating the same, now unphysical, linear map and produces the folded sheets seen at $a = 5$. In contrast, the leapfrog integrator evolves particles under a computed gravitational force, so once particles cross, they are decelerated and cluster into structure. This causes them to oscillate and wind into a bound spiral phase space. This is consistent with the caption of Figure 7.11 of the lecture notes, where the Zel'dovich-predicted structures are noted to disperse again in the absence of gravitational feedback, whereas the true N-body dynamics continue to evolve hierarchically. The spiral in the leapfrog output and its absence in the ZA, therefore provides evidence that the leapfrog integrator is correctly capturing self-gravitating dynamics rather than reproducing the linear approximation with additional computational steps.

Notably, some points of the two solutions overlap at $a = 5$. This occurs where a given particle lies at a phase-space location that both the unphysical linear extrapolation and the true non-linear trajectory pass through simultaneously. However, this does not indicate any deeper agreement between the two solutions beyond the point of shell-crossing.

3.3 PM Force Resolution Check

To check this, `calc_acc_sorted_1d(X)` was implemented, following Chapter 7.2.1 and `calc_acc(X)` from Program Code 7.1 of the lecture notes. In one dimension, the Poisson equation reduces from a partial differential equation to an ordinary differential equation in the single spatial coordinate, which can be integrated directly (Equation 7.53 and 7.54 of the lecture notes). Sorting the particles by position allows this integral to be evaluated exactly and in closed form at each particle location, using only the particle's rank rather than an explicit force calculation (Equation 7.55 of the lecture notes). This results in the exact Newtonian acceleration, with no smoothing or grid-resolution limitation and crucially, it requires no force mesh, unlike the NUFFT calculation used in this work.

To use this computed acceleration within the three-dimensional framework established in `run_simulation()`, the function `calc_acc_exact_3d(X, N)` was implemented, which applies `calc_acc_sorted_1d(X)` independently along each line of particles parallel to the x-axis, placing the result on the x-component of a three-dimensional acceleration array. Meanwhile, the y- and z-components

are set to zero, consistent with the plane-wave initial conditions used, which vary only along x. This is valid because the single-mode plane wave initial conditions of subsection 3.2 have no y- or z-dependence, so each line of particles along x evolves as an independent one-dimensional system. This construction matches the output format of `calc_acc_nufft()` and can therefore be used directly as replacement. Furthermore, `run_simulation_exact()` was obtained by copying `run_simulation()` and replacing only the acceleration function, from `calc_acc_nufft()` to `calc_acc_exact_3d()`, while reusing the rest of the established leapfrog framework unchanged.

`run_simulation()` and `run_simulation_exact()` were then called with the one-dimensional plane wave initial conditions described in subsection 3.2, $L = 1$ and varying particle/grid resolutions for $a_{\text{end}} = 3$. Figure 3 shows the results zoomed in on the central core of phase space ($x \in [0.2, 0.3], v \in [-0.2, 0.2]$). On large scales, both solvers trace the outer spiral reasonably well across all resolutions. However, in the central region where multiple streams overlap and caustics form, a clear resolution dependence appears. At low grid resolution ($N = 32$), the NUFFT force is heavily smoothed over grid cells of width $h = 1/N$, causing it to miss the sharp acceleration gradients at the caustics and fail to wrap the inner core tightly. As N increases, the grid spacing shrinks and the NUFFT trajectory resolves finer scales and matches the exact spiral almost perfectly at $N = 256$.

This behaviour is expected and directly reproduces the resolution dependence shown in Figure 7.9 of the lecture notes; because the NUFFT approach computes gravity via a finite Fourier grid of spacing $h = 1/N$, it cannot represent structure finer than this grid scale and acts as an effective smoothing of the true gravitational force below scales of order h. As the grid resolution increases, this smoothing scale shrinks and the NUFFT force converges towards the exact Newtonian force.

A limitation of the test is that the number of particles is the same as the grid dimension. Consequently, at lower N, the sparser appearance of the exact solution is due to coarse phase-space sampling rather than force inaccuracies. Decoupling the two by fixing $N_{\text{particles}}$ to a high value while varying only N_{grid} would isolate the force mesh smoothing more cleanly. Nevertheless, this comparison quantifies a fundamental limitation of the NUFFT-based approach used for the full 3D simulations in this work; any structure that has collapsed to a scale below the force-mesh resolution will not be gravitationally resolved.

3.4 Time of Shell Crossing

Lastly, the evolution of the minimum determinant of J was investigated for the same single-mode plane wave as in subsection 3.2 and for a Gaussian random field initial condition with $L = 1$, $N = 64$ and $\alpha = 1.5$, to validate the shell-crossing calculation.

Figure 4 shows the results. As expected, for the plane-wave initial conditions,

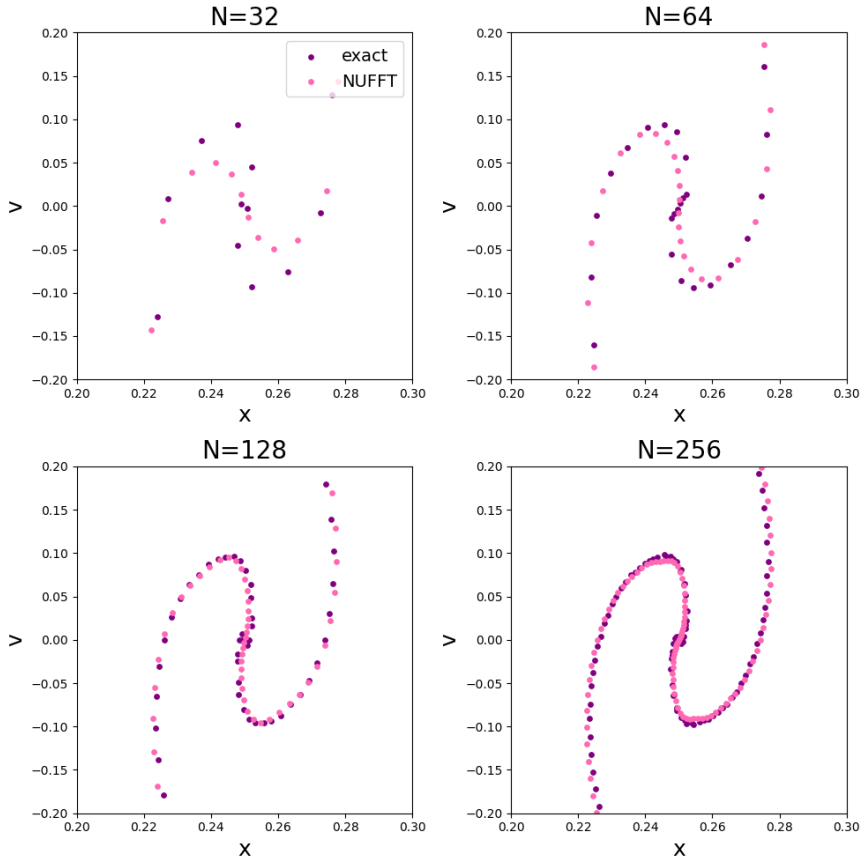


Figure 3: Zoomed-in single-mode plane wave at $a = 5$ for different particle/grid resolutions $N \in \{32, 64, 128, 256\}$ for the NUFFT solver (pink) compared to the exact solution using a sorting algorithm (purple).

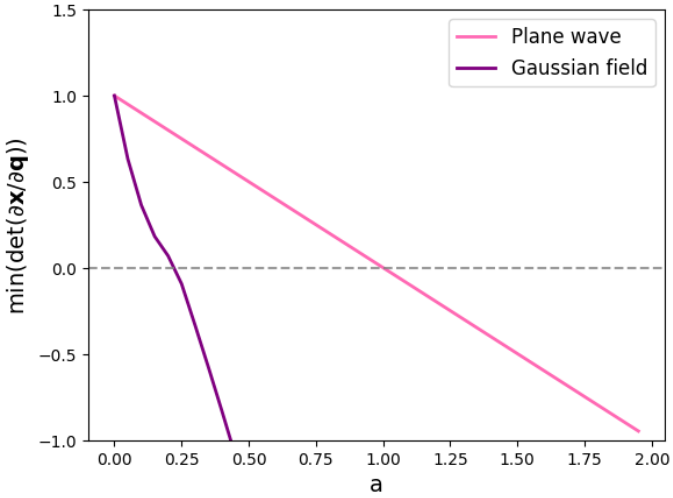


Figure 4: Evolution of the minimum value of the determinant of $\partial\mathbf{x}/\partial\mathbf{q}$ with increasing time a for a 1D single-mode plane wave (pink) and a gaussian field (purple). The dashed grey line marks the condition for shell crossing.

shell-crossing occurs at $a \approx 1$ by construction of the chosen displacement amplitude (subsection 3.2). For the 3D Gaussian field, shell-crossing occurs much earlier, at $a \approx 0.25$.

This earlier crossing time for the Gaussian field is expected, as it contains a broad range of scales, including small scales with large second-derivative amplitudes, compared to the plane wave, which has power concentrated in a single, large-scale mode. Since $\min_{\mathbf{q}} \det(J)$ is sensitive to the most extreme point anywhere in the box, the smallest-scale, highest-curvature fluctuations in the Gaussian field trigger collapse much before any large-scale structure. This figure therefore validates two distinct aspects of the implementation simultaneously; for the plane wave, it confirms that `calculate_shellcrossing()` correctly recovers the known, analytically-predicted crossing time of $a^* \approx 1$ and for the Gaussian field, it confirms that the same code generalizes correctly to a field with a broad range of scales and correctly identifies the earlier first-crossing time.

4 Comparison of ZA and Leapfrog

After the checks above confirmed correct performance of the individual components, the code was used to run full 3D N-body simulations and compare their evolution to that predicted by the ZA alone.

4.1 Visual Comparison

To compare the evolution of a field in time, two simulations were set up from the same initial field using `run_simulation()` with $L = 1$ and $N = 64$. One simulation uses ZA to evolve the field in the linear regime before shell-crossing, followed by the leapfrog integrator after and the second simulation uses purely ZA evolution throughout.

Figure 5 shows the evolution of individual particle positions in a z-slice over time. At $a = 0.1$, particles remain close to their initial grid positions, with only small displacements. Comparing the two columns, the panels appear visually similar for $a < 1$; however, they are only exactly identical at $a = 0.1$, since the shell-crossing time for this field is $a^* = 0.24$, and the switch to the leapfrog integrator occurs at $a_{\text{switch}} = 0.5a^* = 0.12$. At $a = 0.3$ and $a = 0.5$, the leapfrog integrator has already taken 213 and 296 steps, respectively, despite the particle distributions still appearing visually comparable to the pure-ZA case. The first clearly visible difference occurs at $a = 1$, where the ZA panel shows fuzzier edges and fewer overlapping particles than the leapfrog panel. At later times this effect becomes more pronounced; by $a = 5$, particles in the leapfrog simulation have collapsed into compact, well-defined halos, while in the ZA panel they remain much more diffusely distributed. Although the locations of structure agree well between the two panels, the ZA particles do not cluster as strongly, since the Zel'dovich approximation applies no gravitational force after the initial displacement is set and particles simply continue to stream along their initial linear trajectories. This is particularly visible comparing $a = 3$ and $a = 5$. At $a = 3$ the ZA structure still appears reasonably compact but by $a = 5$ the particles have continued streaming past one another and dispersed into a less compact configuration, whereas the leapfrog particles become progressively more clustered over the same time interval.

To compare the fields more clearly, the particle distributions were interpolated onto a regular grid using `cic(X, Y, Z, ngrid)`, which takes the particle positions along the three axes and the number of grid cells per dimension as arguments. It uses Cloud-in-Cell (CIC) interpolation, a multilinear assignment scheme, to distribute each particle's mass among the $2^3 = 8$ nearest grid cells according to its fractional overlap with each (following Equation 7.72 of the lecture notes). After computing the resulting density field ρ on the grid, the function returns the density contrast,

$$\delta = \frac{\rho}{\bar{\rho}} - 1, \tag{24}$$

where $\bar{\rho}$ is the mean density over the box. Compared to the particle scatter of Figure 5, this has the advantage of directly showing the estimated density field itself rather than individual particle positions. This makes the amplitude and sharpness of collapsed structures easier to compare quantitatively through the colour scale.

A visual comparison of the evolution of a slice of the density field over time is

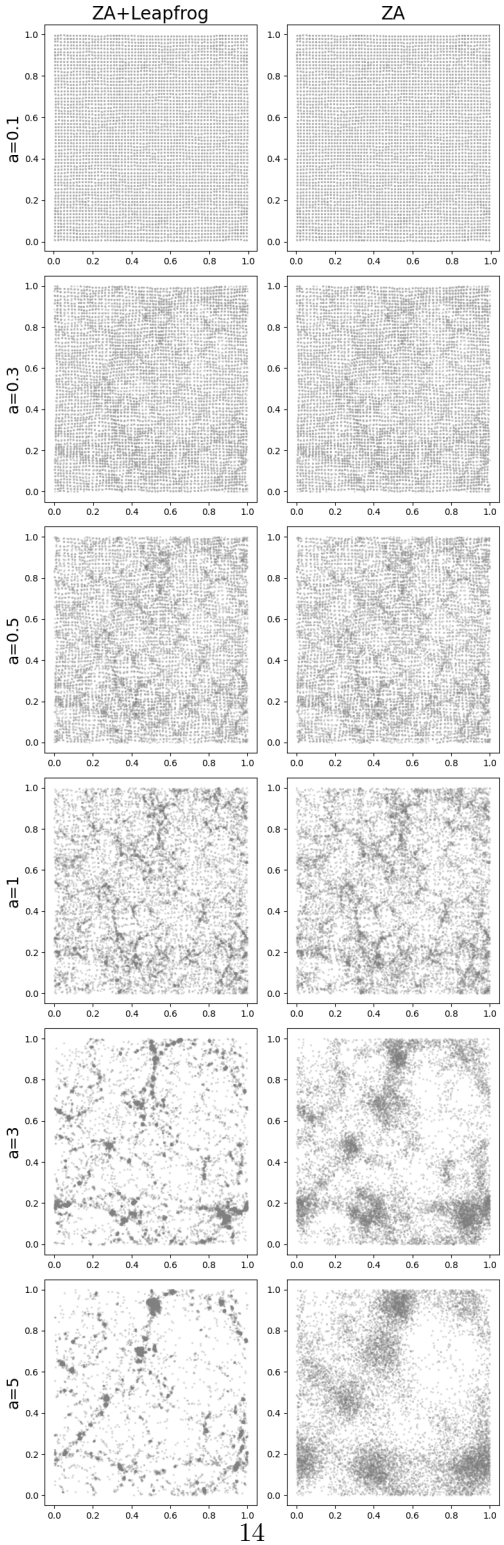


Figure 5: Scatter plot of the evolution of particle positions of an initial gaussian field set up with $L = 1$ and $N = 64$ for time $a \in [0.1, 0.3, 0.5, 1, 3, 5]$. The left column shows the evolution of the particles using ZA in the linear regime before shell crossing and a leapfrog integrator after $a = 0.5a^*$, the right column shows the evolution using purely ZA for a slice through the z-axis at the box center.

shown in Figure 6. Analogous to the scatter results of Figure 5, the fields appear similar until $a = 1$, where the ZA panel begins to show visibly blurrier structure. Comparing the colours, the centers of structures in the leapfrog panel appear brighter, indicating a larger density contrast and therefore more compact and more strongly collapsed structure. This trend becomes more pronounced in the remaining rows, where sharp filaments and halos emerge clearly in the leapfrog panel, while the corresponding structures in the ZA panel remain comparatively diffuse and low-contrast.

Figure 7 shows the evolved fields at the final snapshot $a = 5$ in 3D. This visualization further highlights the difference in clustering; the leapfrog-evolved field (upper panel) shows compact, well-defined halos, whereas the pure-ZA field (lower panel) shows diffuse structures with a substantially lower peak density contrast.

4.2 Standard Deviation

Figure 8 shows the evolution of the standard deviation σ of the density field over time. The standard deviation is a simple measure of the overall amplitude of density fluctuations in the box and a proxy for how far the field has departed from a smooth, unperturbed initial state. While the two curves overlap almost perfectly at early times, the standard deviation of the leapfrog-evolved field grows sharply after shell-crossing, while that of the pure-ZA field grows more slowly and eventually plateaus.

This is expected; prior to shell-crossing, the two methods are identical and σ grows according to linear theory for both. After shell-crossing, ZA has no mechanism to continue increasing the density contrast through gravitational collapse, so the mild growth and eventual plateau of σ in the pure-ZA case reflects only the streaming of the displacement field, rather than ongoing collapse. In contrast, under leapfrog particles continue to gravitationally attract each another after crossing and drive genuine non-linear growth of δ in collapsing regions. This is reflected in the much steeper rise of σ. This figure therefore is as a compact quantitative summary of the qualitative visual comparisons of Figure 5 and Figure 6. It directly demonstrates that the leapfrog integrator captures genuine non-linear structure growth beyond shell-crossing, which the linear ZA is unable to reproduce.

4.3 Power Spectrum

To investigate how well each integrator preserves or generates structure as a function of scale, the power spectra were computed using `compute_powerspectrum3d()` at several snapshots for $L = 1$ and $N = 64$.

Figure 9 shows the results. For both integrators, the amplitude generally increases with time due to gravitational growth. At early times before shell-crossing the curves overlap almost perfectly across all scales, confirming that

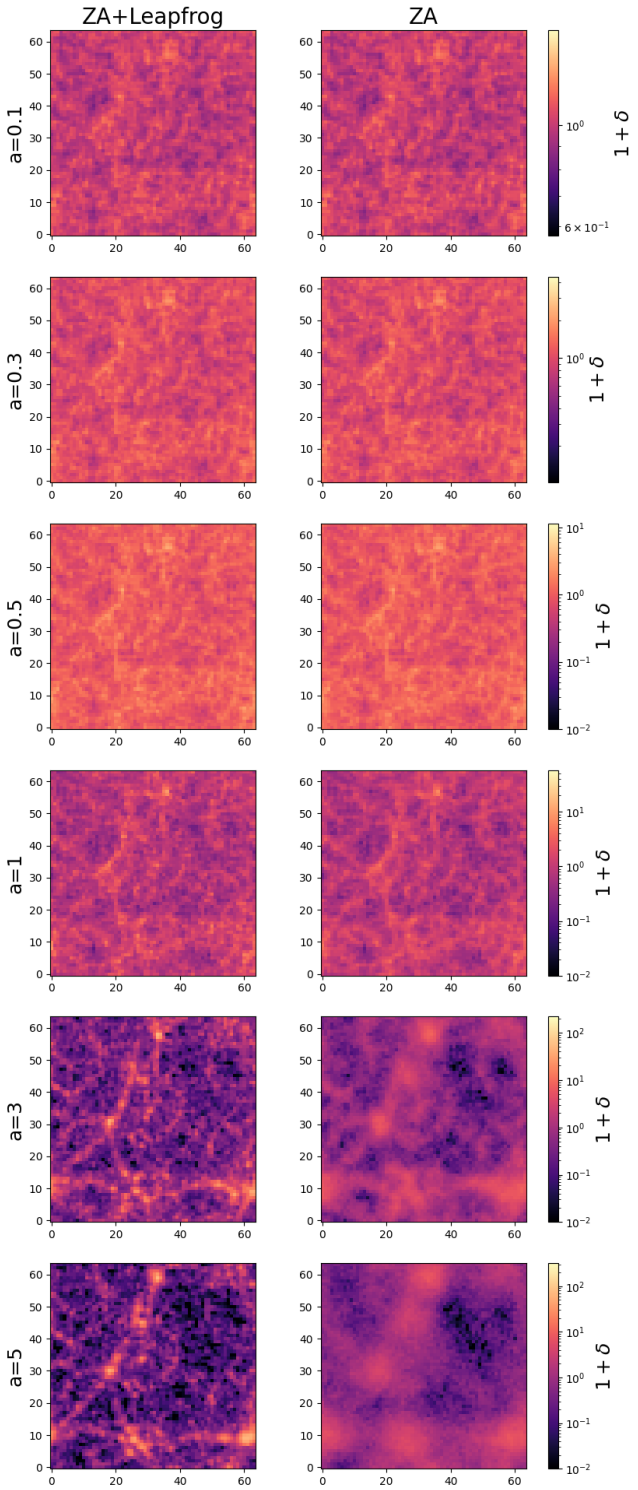


Figure 6: Evolution of an initial gaussian field set up with $L = 1$ and $N = 64$ for time $a \in [0.1, 0.3, 0.5, 1, 3, 5]$. The left column shows the evolution of the field using ZA in the linear regime before shell crossing and a leapfrog integrator after $a = 0.5a^*$, the right column shows the evolution using purely ZA for an averaged slice through the center of the z-axis. The points are colored by $1+\delta$.

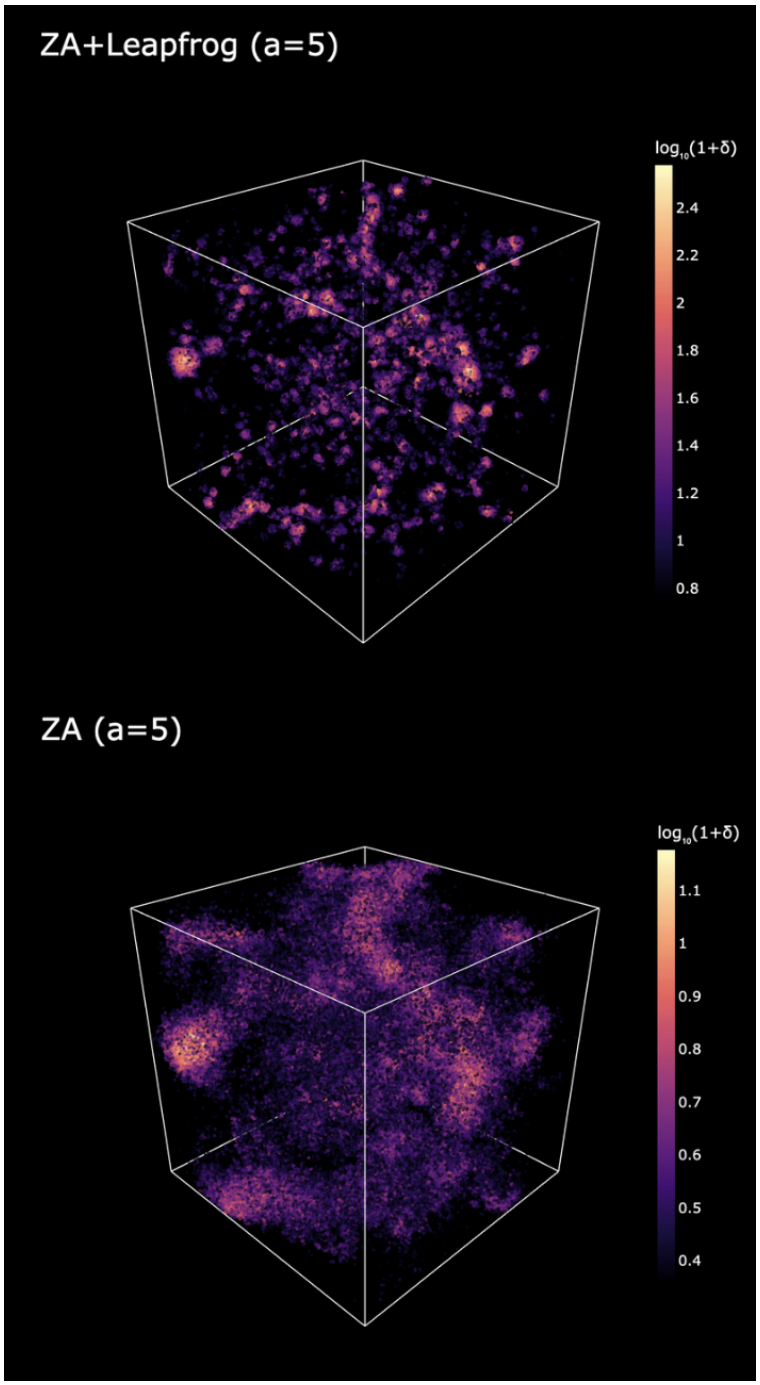


Figure 7: Comparison of the evolved fields at $a = 5$ in 3D; ZA+Leapfrog (top) and pure ZA (bottom).

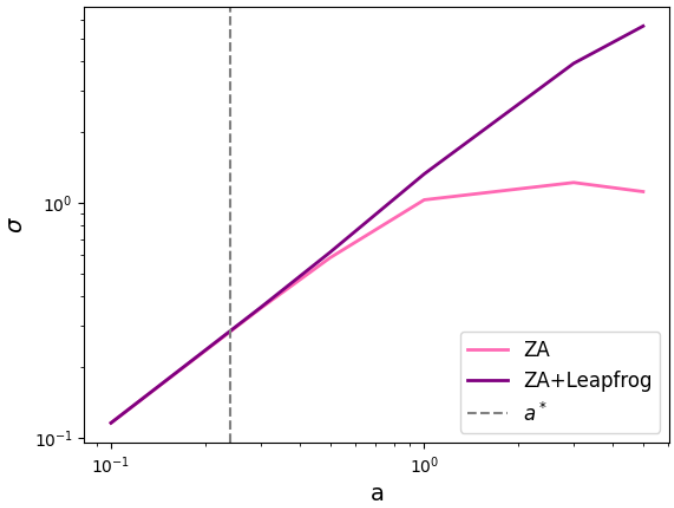


Figure 8: Evolution of the standard deviation σ of the field with increasing scale factor a for the ZA+leapfrog simulation (purple) and the pure ZA simulation (pink). The dashed gray line marks the time of shell crossing a^* for this simulation set up.

both integrators remain in the linear regime, where ZA is exact. By $a = 0.5$, the power spectrum of the ZA-evolved field begins to fall below the leapfrog result at small scales, as small scales are the first to evolve non-linearly, while large-scale modes remain in the linear regime for longer. As a increases further, this deviation extends to larger scales, showing that non-linear evolution starts to dominate an increasing range of scales over time. By the latest snapshot $a = 5$, the ZA power spectrum falls below the leapfrog result across the full range of scales shown. This demonstrates quantitatively, the same conclusion drawn visually above; the ZA approximation increasingly under-produces clustering relative to the true non-linear evolution as time progresses, with the discrepancy spreading from small to large scales as structure formation proceeds hierarchically.

5 Conclusion

In this work, an N-body simulation code was set up and its output compared against results from the pure linear Zel'dovich approximation.

Validation and performance checks were first constructed to ensure correct functionality before running more expensive 3D simulations. These checks tested individual functions of the simulation code as well as the full simulation pipeline in a controlled one-dimensional setting, where results could be directly compared

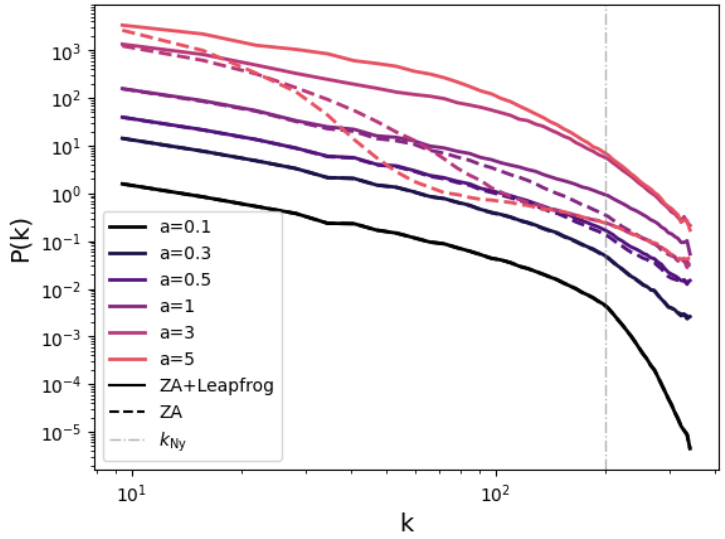


Figure 9: Evolution of the power spectrum with increasing scale factor a for the ZA+Leapfrog simulation (solid lines) and the pure ZA simulation (dashed lines). The Nyquist frequency is marked with a gray dash-dotted line.

to the expected outcomes described in the lecture notes. First, the power spectrum encoded in the initial Gaussian field was validated by comparing its shape to the theoretical prediction. The evolution of a single-mode plane wave with increasing scale factor then produced the expected qualitative behaviour for both integrators; the Zel'dovich approximation produced folded, unphysical sheets in phase space past shell-crossing, while the leapfrog integrator produced a tight, converging spiral, which is a sign of genuine gravitational collapse. A resolution check of the particle-mesh force further showed that the NUFFT-based solution converges towards the exact Newtonian force as the grid resolution is increased. Finally, the shell-crossing time computed for the plane wave agreed with the analytically predicted value, and for a full Gaussian field, shell-crossing was correctly found to occur earlier, confirming that the shell-crossing detection generalizes correctly beyond the single-mode test case.

The main goal of this work was to compare the 3D evolution under ZA to the full leapfrog integrator using qualitative and quantitative measures. The evolution of particle positions over time showed that the two solutions agree closely at early times but diverge markedly after shell-crossing; the leapfrog integrator produces well-defined halos clustered along filaments, while particles in the ZA case remain significantly more diffuse. Computing the density field via CIC interpolation further revealed much higher density contrast values in the leapfrog results, confirming the formation of compact structures. This was supported quantitatively by the evolution of the standard deviation of the density field,

which grows much more strongly for the leapfrog integrator after shell-crossing, therefore reflecting non-linear structure growth. Finally, the power spectra of both methods computed at several snapshots showed that ZA fails to reproduce power at small scales as soon as they become non-linear after shell-crossing and that this progressively extends to larger scales as time progresses.

This is a basic N-body simulation, which was built and validated from first principles and therefore the accompanying analysis is correspondingly simple. The implementation is also computationally slow relative to production-grade codes.

Several extensions could improve the performance, physical fidelity and diagnostic depth of this implementation. Since the current NUFFT calculation dominates the runtime, separating the force-mesh resolution from the particle count, implementing adaptive local time-stepping, parallelizing or GPU-accelerating the force calculation could drastically increase the speed. Physically, using higher-order Lagrangian initial conditions, as discussed by [Michaux et al., 2020], would allow a later and therefore cheaper switch to the full N-body integration. Moreover, the simple and conservative choice of a_{switch} used here could be exchanged with a more sophisticated criterion or a symplectic Zel'dovich-consistent integrator, which would allow the transition to be delayed further without loss of accuracy. Finally, this work only investigates visual comparisons, the standard deviation of the density field and the power spectrum. A natural extension would be to identify and characterize individual collapsed structures directly, for instance via a simple friends-of-friends or spherical-overdensity halo finder, and to compare the resulting halo mass function between the ZA and leapfrog simulations. This would allow a direct comparison of the abundance and properties of collapsed structures. More generally, running the comparison at higher resolution and in a larger box with initial conditions tuned to reproduce a more realistic power spectrum rather than the simple power-law spectra used here, would allow a more direct, quantitative comparison to results such as those of [Springel et al., 2005].

Nonetheless, the simulation used in this work reproduces the expected physical behaviour at each stage of validation and produces visually pretty results. Overall, the comparison highlights the necessity of switching from the Zel'dovich approximation to a full N-body integrator sufficiently early before non-linear structure formation begins, in order to correctly capture clustering.

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